X4Pro. News-2026.

by V.Zerkin, Vienna, 2026-08-28

1. Code reac1.py: universal retrieval and plotting of EXFOR data having one-string Reaction-codes, for example:
   - *CSP Partial cross section data*
   - *CST Temperature dependent cross section*
   - *ETA Average neutron yield per nonelastic event*
   - *NU Fission-neutron yield, nu-bar*
   - *TKE Total kinetic energy*
   - *SIG Cross section data given with limit as DATA and DATA-MIN*

   -- Currently implemented search: by full-Reaction-code, DatasetID, Author1
2. Plotting partial cross sections from local EXFOR and remote ENDF databases
   - *CSP vs ENDF:MT51:MF3+33 - Production of a neutron, with residual in the 1st excited state*
   - *NU vs ENDF:MT452/455/456:MF1+31 - Average number of total/delayed/prompt neutrons released per fission event*

Code reac1.py help:

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Program: reac1.py, ver.2026-08-14
Author:  V.Zerkin, Vienna, 2021-2026
Purpose: Retrieve and plot any type EXFOR data from local EXFOR database
Introduction:
  * computational data are presented as function y=f(x1,x2,..)
    where f is measured quantity, xi are independent variables:
      EN  - incident energy,                  E2  - energy of outgoing particle,
      ANG - angle of outgoing particle,       EVL - energy level, etc.
Algorithm:
  * user defines EXFOR Reaction-code for data search, x-variable by number
  * selects x-variable by number (default: x1)
  * other variables will be used as parameter for grouping of data points
Usage:
    $ python [{flag}] x4update.py {[option|reaction]}
      * flag: see all Python flags:  $ python --help
        -B        don't write .pyc files on import
      * option:
        -help       print this help-text and exit (also --help)
        -o:<file>   output file name
        -x:x?       set x variable: "x1" to "x5", e.g. -x:x2, default: -x:x1
        -x?fam:<str> filter for "families" of x? variable, e.g. -x2fam:"LVL;EXC"
        -x?min:<num> filter out values of x? variable, e.g. -x1min:6e6
        -x?max:<num> filter out values of x? variable, e.g. -x1max:30e6
        -nogrp      avoid groupping datasets by reaction-codes on the plot
        -fx:<num>   set factor for x-values, e.g. -fx:1e6 sets "MeV" on X-Axis
        -fy:<num>   set factor for y-values, e.g. -fy:1e-3 sets "mb/sr" on Y-Axis
        plotting:
        -xlog       logarithmic scale of X-Axis
        -ylog       logarithmic scale of Y-Axis
        -lines      connect points on the plot by lines
        -sym        draw border of a symbol of a data point on the plot
        -annot:x,y,txt  annotation on top of the plot;
                        x and y should be given in units of the plot;
                        text can include html tags: <b>, <sup>
      * reaction:
        <reaction>  full EXFOR Reaction-code, e.g. "8-O-16(N,EL)8-O-16,,DA"
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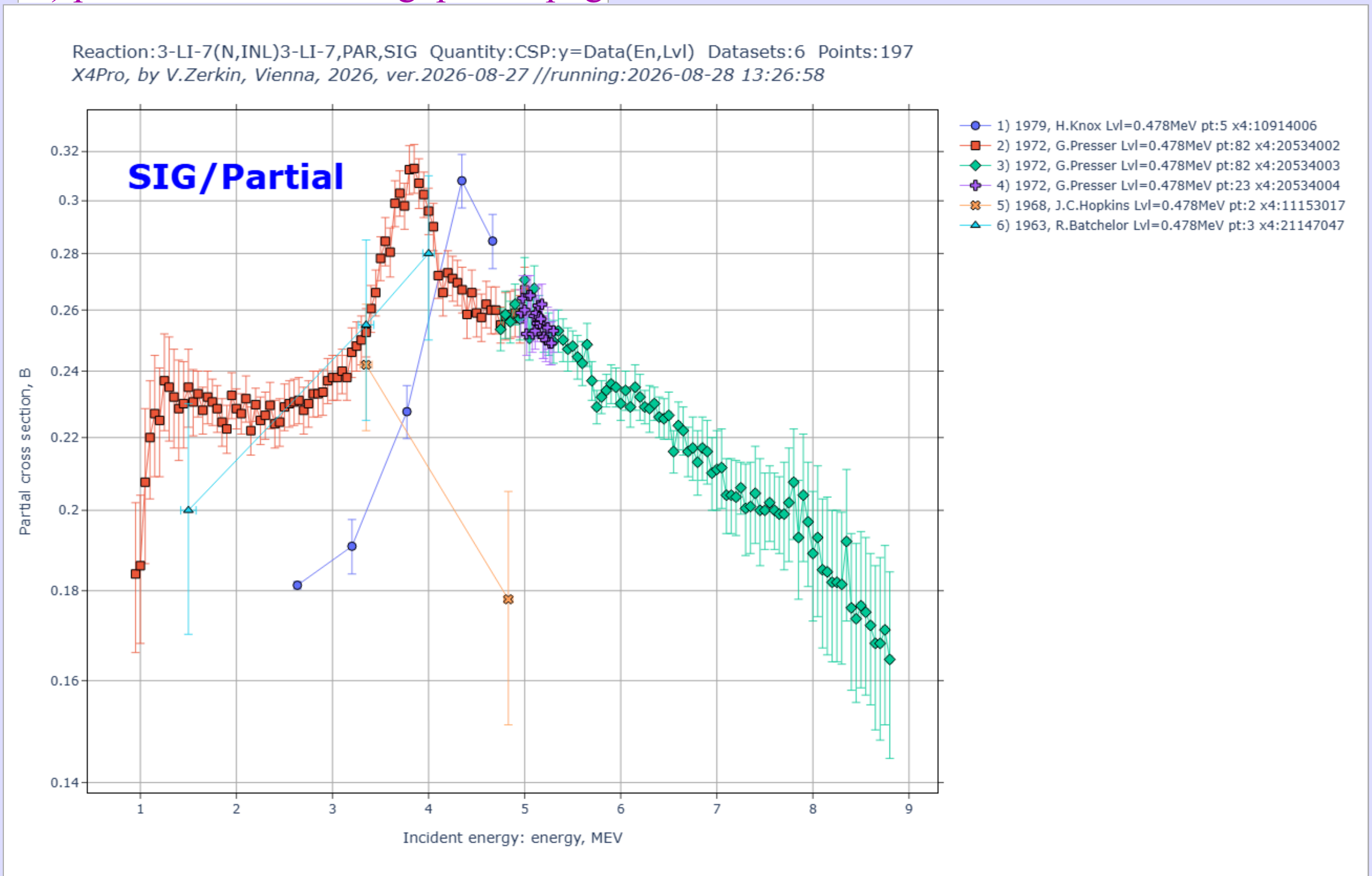
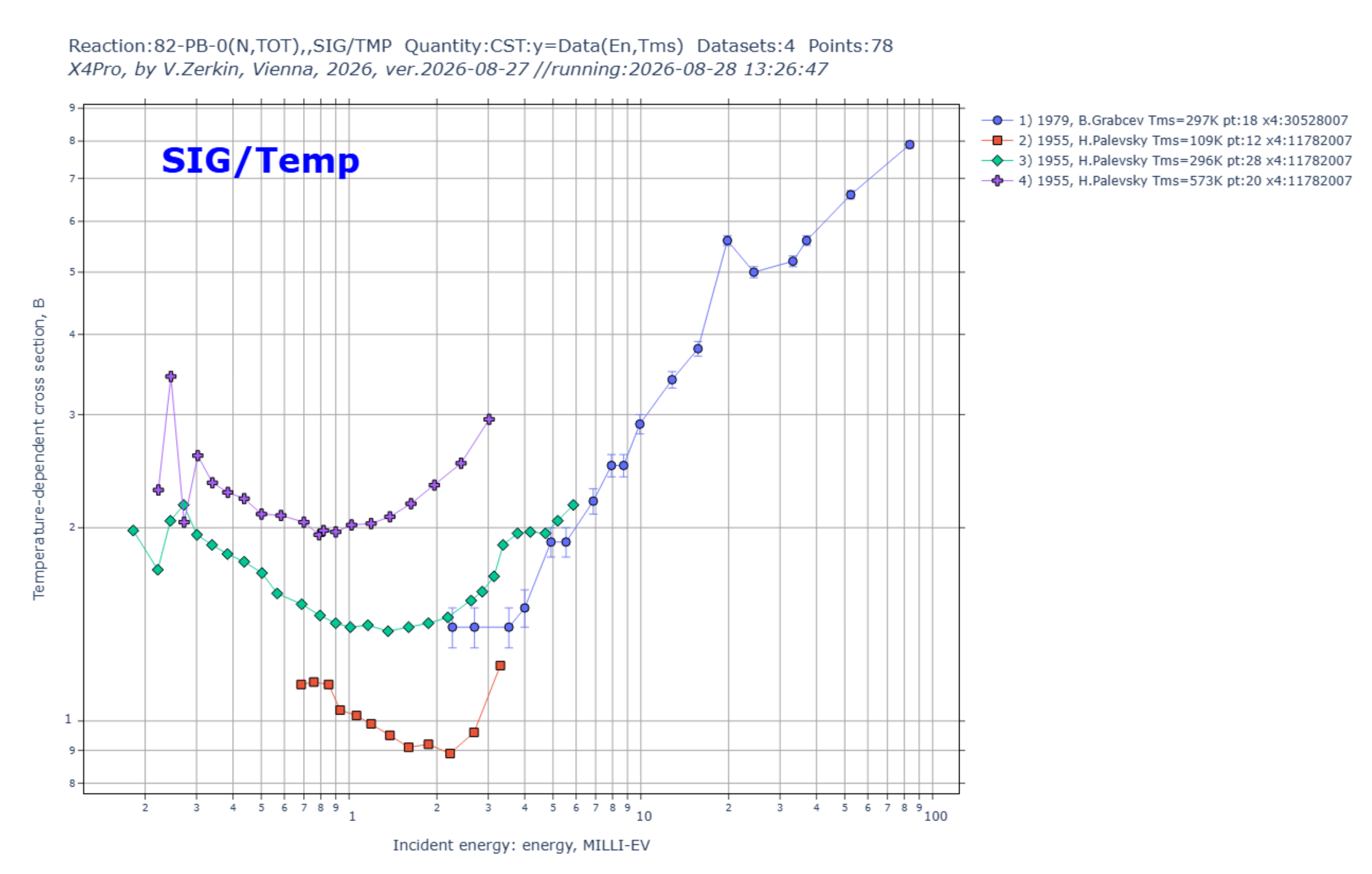
part1-8-reac/ *EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA*

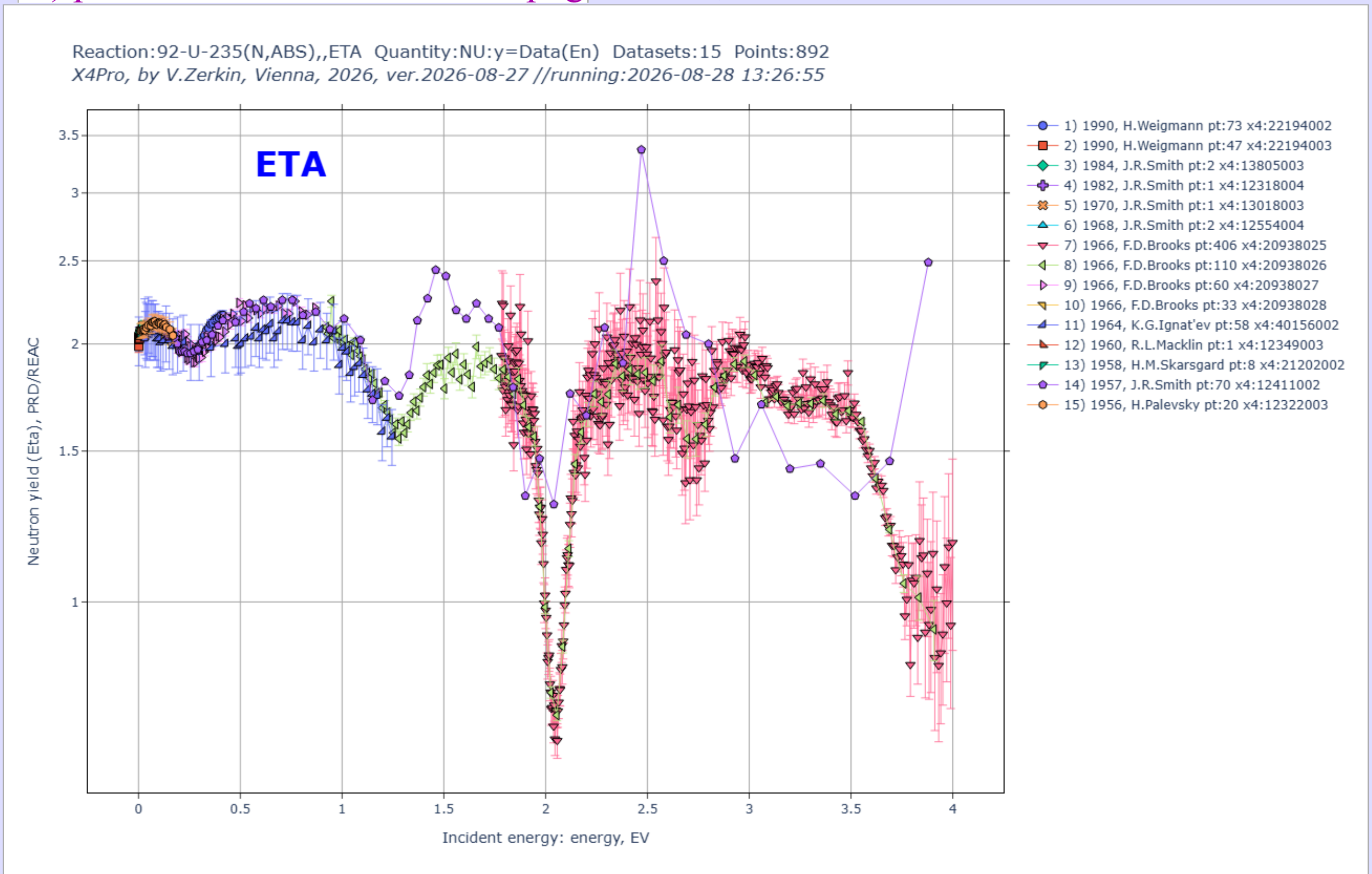
1) part1-8-reac/out00/sig1p.html.png CSP Partial cross section data



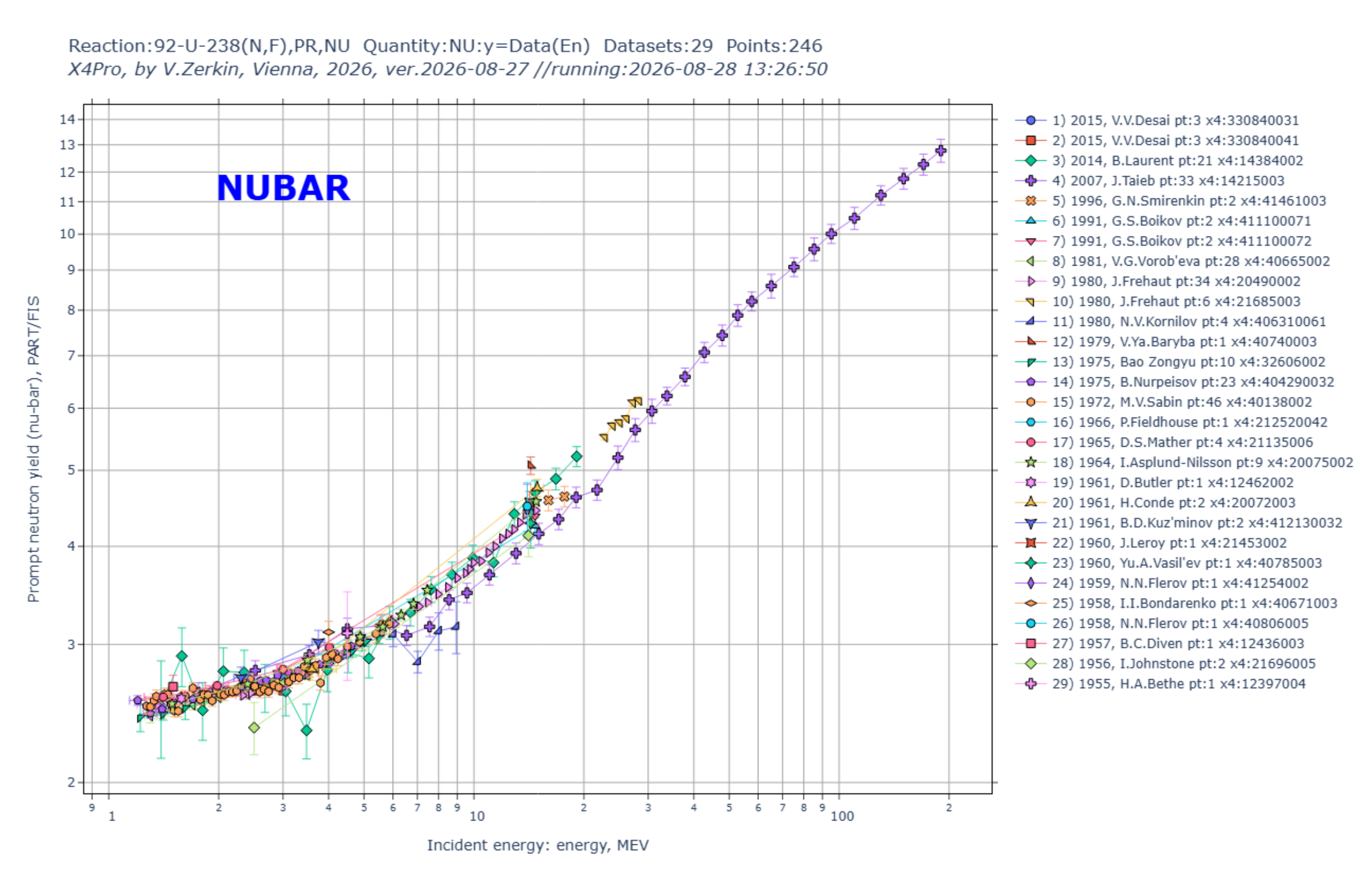
2) part1-8-reac/out00/cst1.html.png CST Temperature dependent cross section



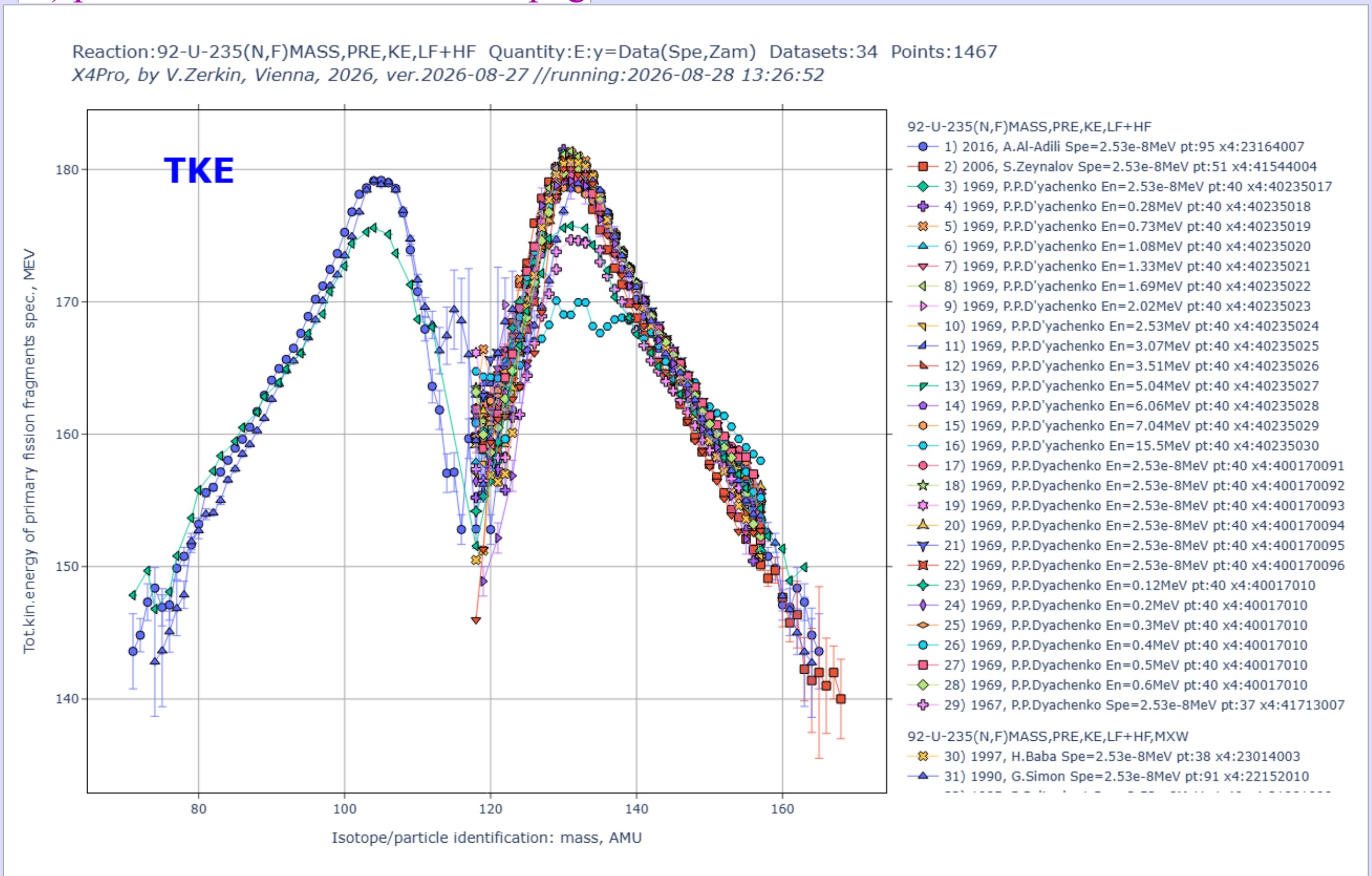
3) part1-8-reac/out00/eta1.html.png ETA Average neutron yield per nonelastic event



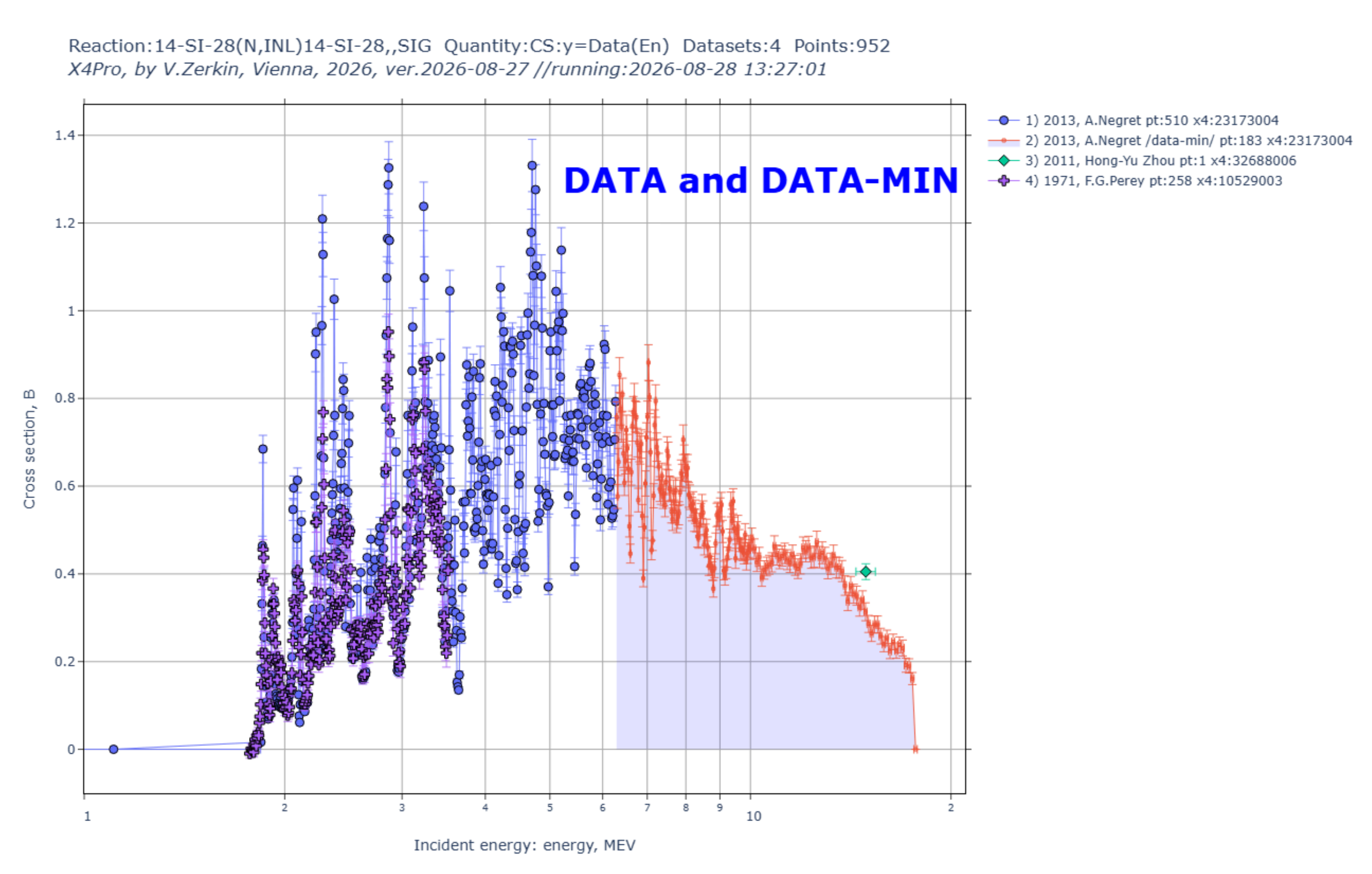
4) part1-8-reac/out00/nu1.html.png NU Fission-neutron yield, nu-bar



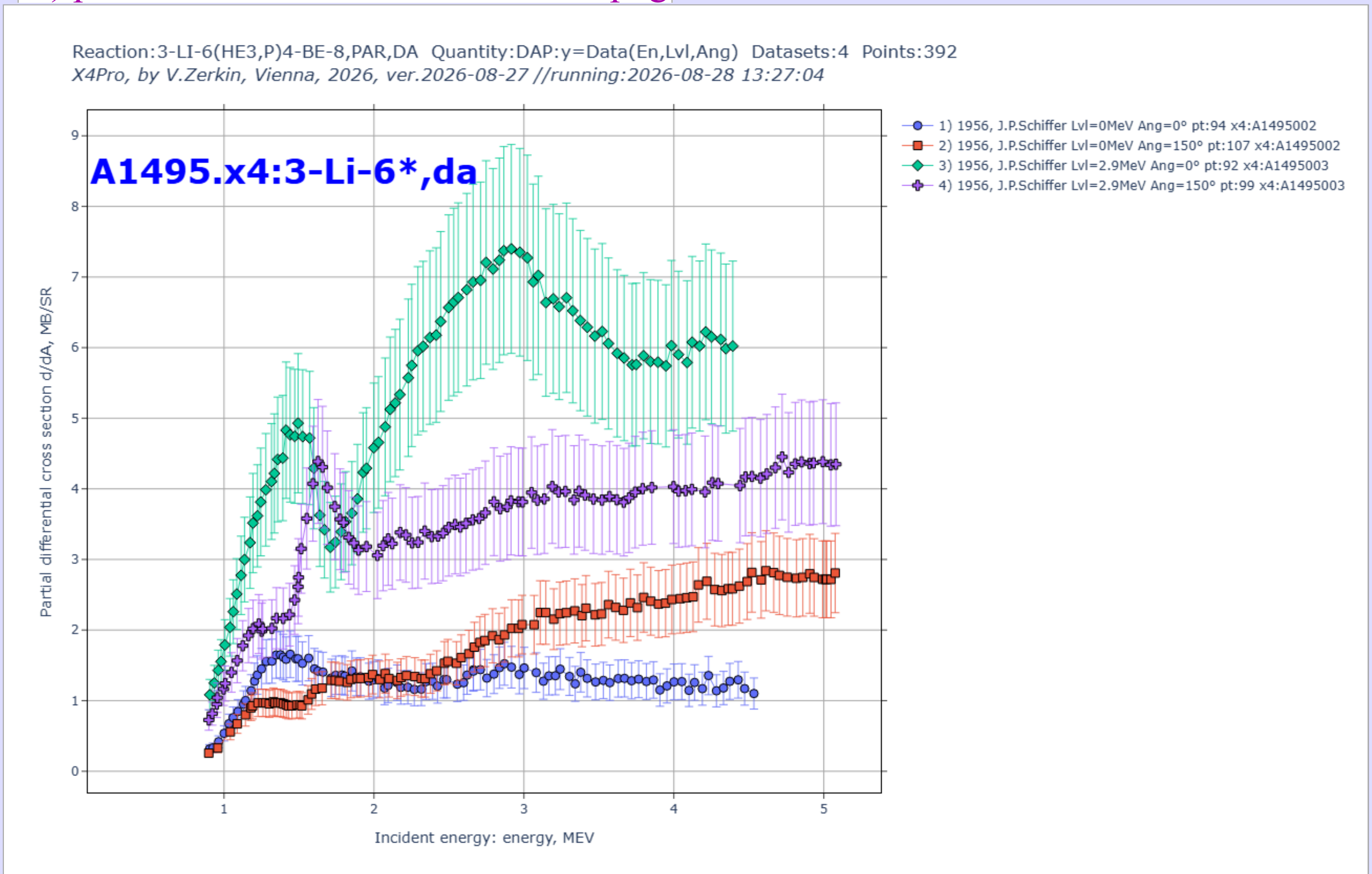
5) part1-8-reac/out00/tke1.html.png TKE Total kinetic energy



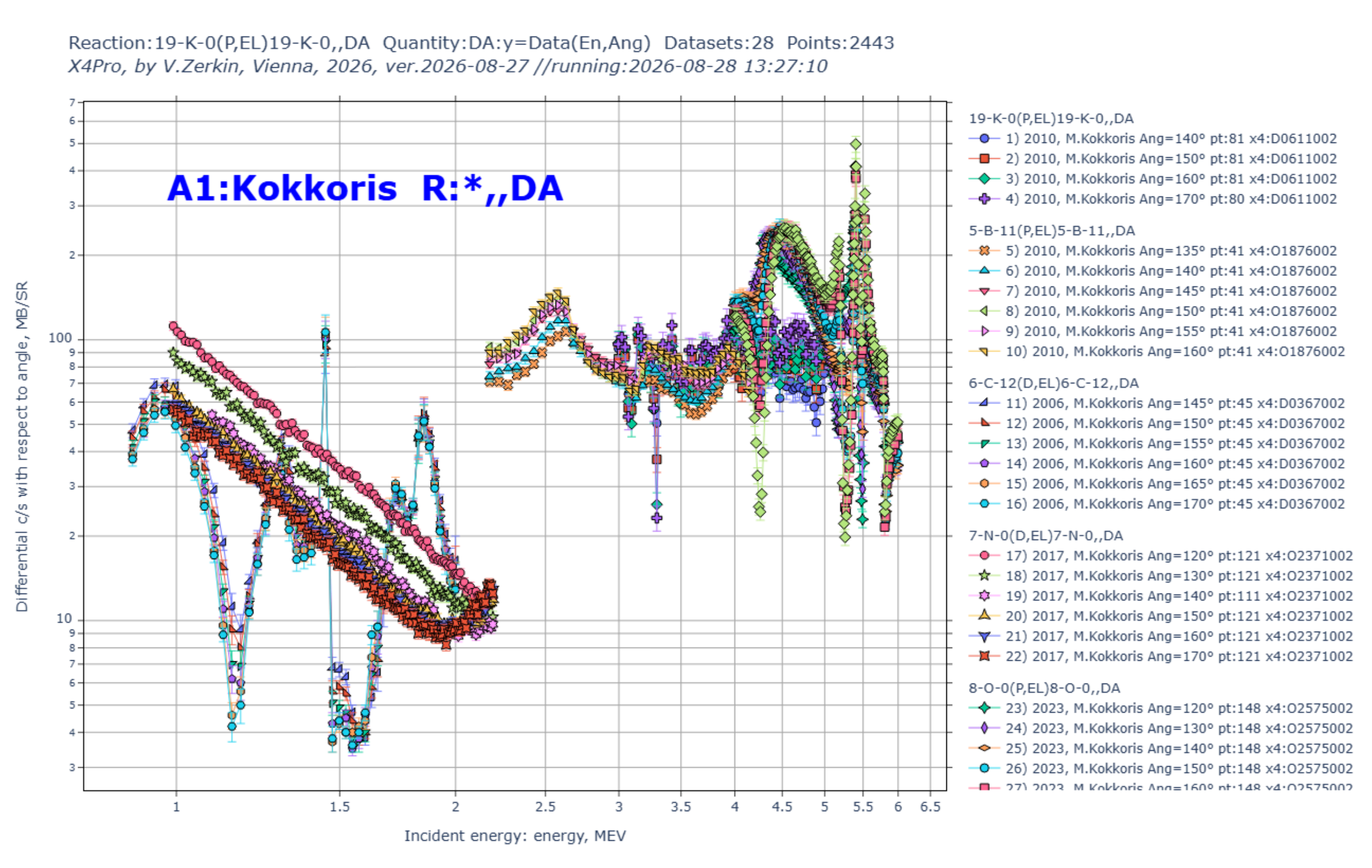
6) part1-8-reac/out00/si28nn.html.png
SIG Cross section data given with limit as DATA and DATA-MIN



7) part1-8-reac/out00/a1495-li6.html.png
Data from EXFOR:"A1495" & Reaction: "3-LI-6*DA"

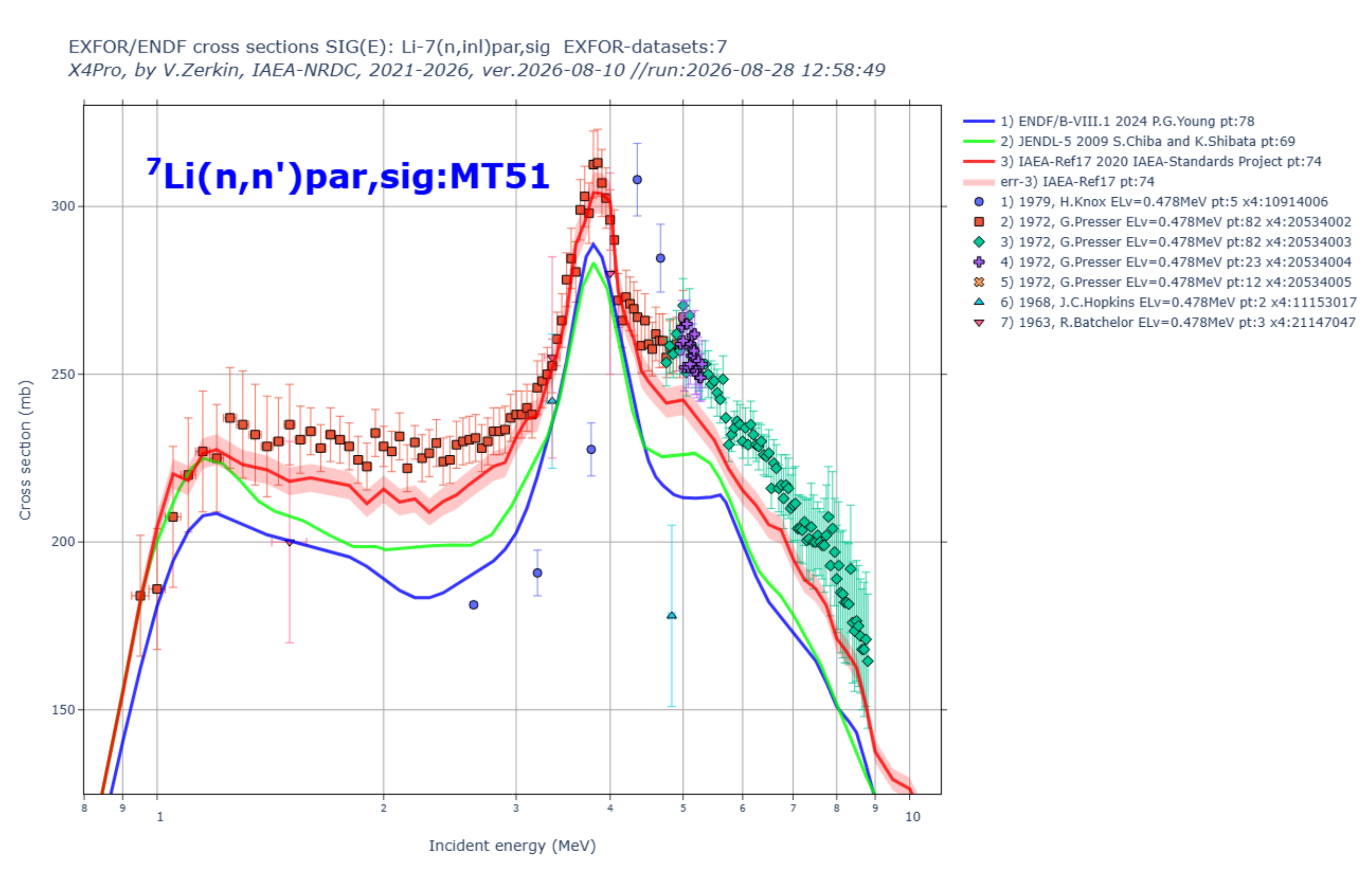


8) part1-8-reac/out00/Kokkoris.html.png Data by "Kokkoris" & Reaction: "*,,DA"



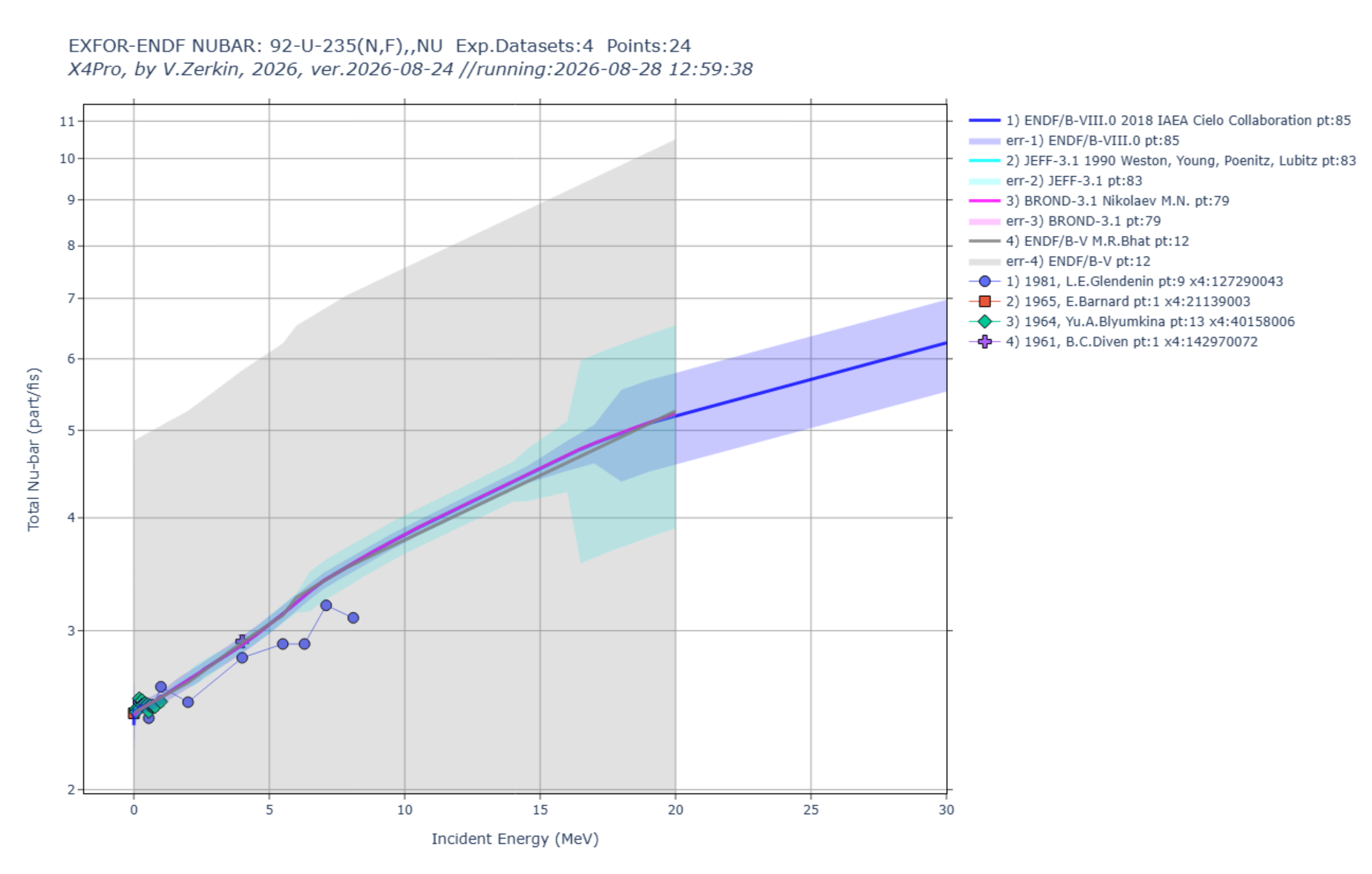
part2-8-sig1par/ *EXFOR/ENDF partial cross sections CSP:MT51*

9) part2-8-sig1par/out00/Li7ninl.html.png Production of a neutron, with residual in the 1st excited state

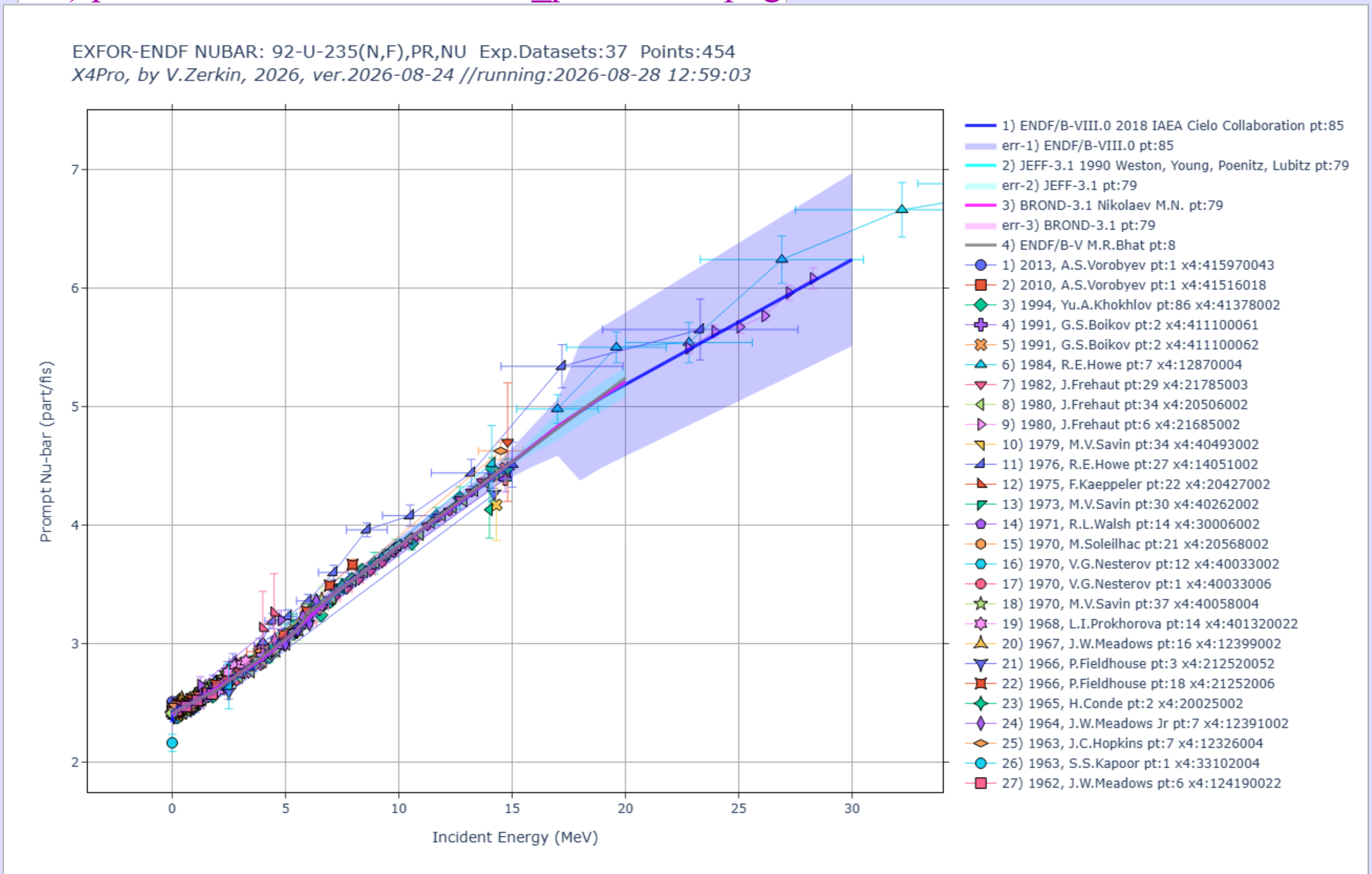


part2-9-nubar/ *EXFOR/ENDF NUBAR - average number of neutrons per fission*

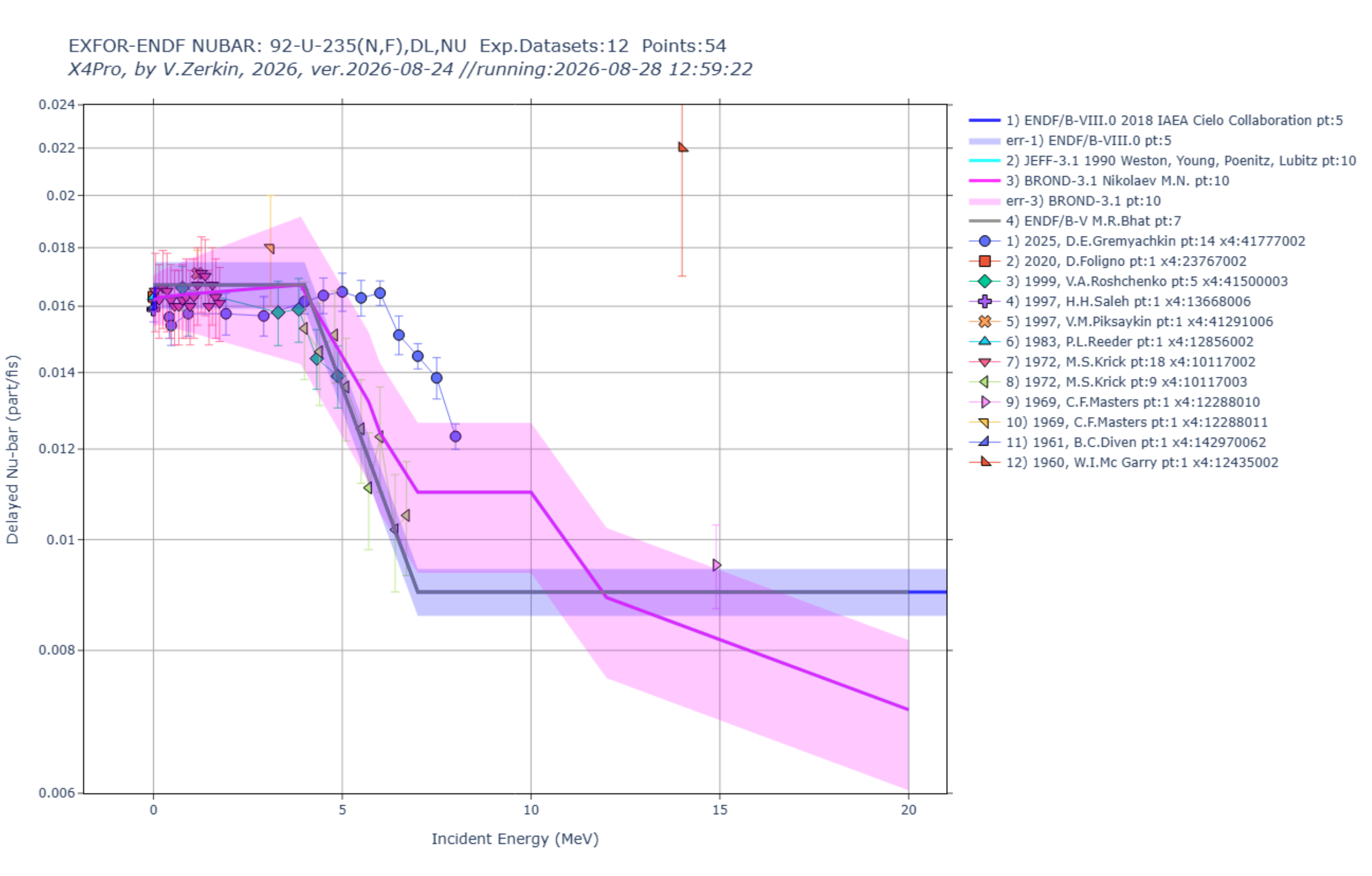
10) part2-9-nubar/out00/U235nf_-nu.html.png Total neutron yield (nu-bar)



11) part2-9-nubar/out00/U235nf_pr-nu.html.png Prompt neutron yield (nu-bar)



12) part2-9-nubar/out00/U235nf_dl-nu.html.png Delayed neutron yield



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